



Lahore 水与废水管理项目 (LWWMP)
AIIB 项目编号 P000074 合同编号 0074-PAK-01

第三套澄清文件

采用设计、采购和施工模式 (EPC/交钥匙方式) 的工程单阶段双信封采购

LAHORE 废水和排水管理项目 (LWDMP) - 从 LARECHS COLONY 至 GULSHAN E RAVI

的污水管网 (采用非开挖技术)

2025年12月17日



ASIAN INFRA
INVESTMENT BANK

Lahore 水与废水管理项目（LWWMP）从 LARECH COLONY 至 GULSHAN-E-RAVI
Lahore 的污水管网投标人询问事项第三套澄清文件

Sr #	Reference Section	Description	Employer's Response
1.	Volume 1, Part I, Section 4, Tender Forms Page 4-46-4-62 Schedule No. 5: Electrical Works	Electrical works in 23 Intermittent Pumping Stations Electrical works along sewer lines (15 Nos. Control System at Drop Structures) & 15 Nos. Control Points on Manholes along Lines and Laterals preferably at drop structures in accordance with ER Sub Clause 4.1 & 5.15 Nos. (minimum) Control Points on Manholes along Lines and Laterals ULTIMATE DISPOSAL STATION AT GULSHAN E RAVI (ELECTRICAL WORKS FOR PUMPING STATION AND ALL BUILDINGS REFERRED IN EMPLOYER'S REQUIREMENTS The Tenderer notes that only the Single Line Diagram (SLD) for the Disposal Station has been provided. The Tenderer respectfully requests the Employer to provide the load list, SLDs, layout drawings, and any additional electrical drawings for other stations, if available.	It is to clarify that EPC Contractor is responsible for detailed design of loads and SLDs for all the stations, however, the load requirement for all 23 intermittent pumping stations and typical SLDs for odor control system, ventilation system and screening system is attached as Annex-A for guidance only.
2.	Volume 2 Part II, Section 6, Employer's Requirements	It is recommended that the panel rooms can be adjusted in the dry well chamber. Can this alternate arrangement be acceptable to the Employer.	Kindly refer to response at Sr. No. 39 of the Clarifications and Minutes of Meeting of 1st Pre-Tender Meeting, wherein, the updated drawing # LW-TD-411 shall prevail, however, as an alternative arrangement, the panel rooms can be adjusted in the dry well chamber as per Annex-B.
3.	Clarifications and Minutes of Meeting of 1st Pre Tender Meeting LWDMP Item 72 Volume-1, Part I, Price Schedules, Schedule No. 5: Electrical Works, Item No. 5.6.	It is to clarify that shifting of existing Generators and Control rooms at 23 lift stations at new location (within the existing premises) will require Medium Voltage Cables for restoring / reinstating the connection with the existing Pumping system. In accordance with the Employer's requirements for Medium Voltage (MV) cables to restore the connection with the existing pumping system, the Tenderer requests clarification on the following: 1) Please provide the load information, detailed load	1) The EPC contractor is at liberty to visit the relevant site and estimate the relocation and reconstruction scope of work, as per its own design and layout. Further, refer to response of Item 72 of Clarifications and Minutes of Meeting of 1st Pre Tender Meeting LWDMP, it is to clarify that 23 pumping stations other than Gulshan e Ravi pumping station have LV loads and these pumping stations will require Low Voltage Cables for restoring / reinstating the connection with the existing Pumping system, if shifting involves.

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	4, Tender Forms Page 4-46 Schedule No. 5: Electrical Works Volume 2, Part II, Section 6, Employer's Requirement	16 control rooms & 28 buildings/ rooms reconstructed & 14 generators operated & 11 control panels shifted approximately in accordance with ER Sub Clause 4.1 & 5.1). The Tenderer requests the Employer to provide details for the 14 generators, including voltage rating, capacity, dimensions, and any additional relevant information. Further, please provide a basic design or overview of the Electrical (EL) System covering all stations under this project.	attached as Annex-C. The EPC contractor is at liberty to visit the relevant sites and prepare the design of proposed infrastructure within available premises by keeping existing system intact as per criteria / specifications given in Employer's Requirements. In addition, details regarding available load at 23 lift stations and other electrical installations is made part of Annex-C
6.	Volume 1, Part I, Section 4, Tender Forms Page 4-46 Schedule No. 5: Electrical Works Volume 2, Part II, Section 6, Employer's Requirement	(23 Nos. Intake Structures for 23 Screen Systems & 16 control rooms & 28 buildings/ rooms reconstructed & 14 generators operated & 11 control panels shifted approximately in accordance with ER Sub Clause 4.1 & 5.1). The Tenderer notes that only the 14 generators located at the existing intermittent pumping stations are mentioned in the ITB. Please clarify: 1) Whether the remaining 9 stations require new generators or if they have existing generators that may be utilized. 2) Whether the capacity of the 14 existing generators is sufficient to meet the new stations' load requirements.	It is to clarify that list of Generators available at 23 lift stations is attached at Annex-C, however, as per preliminary design and drawings, shifting of only 14 generators is envisaged, however, the Tenderer is at liberty to avoid shifting during formulation of its design / costing. Further, 1) As per current assessment, new generators are not required at remaining 9 intermittent pumping stations. 2) The capacity of the existing generators is considered to be sufficient to operate the attached loads.
7.	Volume 2, Part II, Section 6, Employer's Requirement xxxix) Design and supply of SCADA System, Page6-73	The SCADA hardware will include: Sewage video cameras and surveillance systems for visual inspection of trunk sewerage lines and disposal stations. According to the referenced clause, please clarify whether video surveillance cameras are required along the entire sewage pipeline route or only at critical facilities.	As responded in Sr. No. 90 of the Clarifications and Minutes of Meeting of 1st Pre-Tender Meeting, it is to clarify that Sewage Video Cameras shall be installed at critical facilities. The cameras will be installed, where ever RTUs are to be installed in the manholes along the sewage pipeline and at critical locations as site requirements.
8.	Volume 2, Part II, Section 6, Employer's Requirement DATA ACQUISITION FOR SCADA AT VARIOUS	a) Remote Terminal Units (RTU) shall be installed at selected (jacking shafts) sites to retrieve the data and transmit through the 4G (GSM/GPRS) media to the new disposal station at Gulshan Ravi. List of RTU sites is provided.	It is to confirm that all RTUs and cameras at pipeline shall communicate using 4G media.

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	SEWERAGE COLLECTION ON MAIN SEWER LINES, Page 6-332	b) Option of wireless radio is recommended as primary media as it is readily installable and cover wide area of communication. Data transmitted per station, in general for this project, is in the range of 1 ~ 5, which is easily manageable with 4G technology with required performance. As per the clause and Volume 2, Part II, Section 6 of the Employer's Requirements (Page 6-323), the Tenderer understands that all RTUs and cameras at pipeline stations will use a leased 4G GSM wireless network for data transmission, and that no new optical fibre cables will be installed. Please confirm this understanding.	
9.	Volume 2 - Part II - Employer's Requirements Sub-Clause 4.1 Contractor's General Obligations and Volume 3-A - Part II -Tender Drawings	Line-A will collect flows from disposal stations of Karachi Phattak/ Harbanspura DS, Main Mir Colony, Infantry Road, Bird Market, Larech Colony, Ghazi Mohalla, Muhammad Nagar, Governor House, Lakshmi Chowk, and Janki Devi. Line-B collect sewage flows from disposal stations Guru Mangat, Q-Block, CentrePoint, G-Block, Zafar Ali Road, Shareef Colony, Ichra, Shadman, Shama, and Rasool Park. Line-C collects sewage flows from disposal stations of Lytton Road, PIA Planetarium, and Poonch Road. The Tenderer notes that no drawings have been provided in V3A - Part II - Tender Drawings for the Harbanspura / Karachi Phattak Disposal Station. Kindly define the scope of works and provide drawings for the inlet chamber, outlet chamber, mechanical screen, drop structure, and odour-control system	Please refer to the reply to Sr. No. 26 of the Clarifications and Minutes of Meeting of 2nd Pre-Tender Meeting, that may be accessed on the WASA Lahore website under LWWMP Tab.
10.	Volume 3-A - Part II - Tender Drawings Drawing no: LW-TD -491 ULTIMATE DISPOSAL STATION SINGLE LINE DIAGRAM (SLD)	In Drawing No: LW-TD -491, there are water supply pumps and fire pumps however, the relevant parameters and requirements for these facilities are not specified in the tender documents. The Tenderer requests clarification on the following: 1) Whether fire water supply facilities are required at the Ultimate Disposal Station. If yes,	1. Kindly refer to Section 6 Employer's Requirements, Environmental & Social Management Plan (ESMP), Table 1-5: Environmental and Social Management Plan during Construction and Operational Phase, it is to clarify that fire water supply facilities are required at the Ultimate Disposal Station and the system will be designed and constructed according to National Fire Protection Association (NFPA)

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		please provide the type and design parameters (flow rate, head) of the fire pumps. 2) Please specify the fire-fighting systems required for each building (fire hydrant system, gas-based fire-extinguishing, automatic sprinklers, etc.).	standards, by using design., codes/references mention hereunder: NFPA 20 (fire pumps), NFPA 13 (automatic sprinklers), NFPA 14/NFPA 24 (hydrants / private mains), NFPA 22 (fire water tanks), and NFPA 2001 (clean-agent systems). Provide a standalone fire water supply capable of supporting private hydrants and sprinklers for a minimum 2-hour duration dedicated water tank at the chosen design flow. System Requirement: Install external hydrants (industrial style) distributed to provide required coverage; hydrant outlets, 2.5-in nozzles, and provisions for fire department connections (FDC). Plant buildings, workshops, stores, hazardous material storage (if any): Hydrant system inside/outside + hose reels as required; consider local sprinklers inside enclosed buildings where combustible materials are stored. Additional: manual fire extinguishers, detection & alarms, emergency lighting, and signage. Jockey pump (pressure maintenance): Jockey pumps are small, typical flow 5-10% of main pump flow to maintain pressure and avoid main pump cycling. Design Pressure: Consider a minimum of 7 barG at the hydraulically most remote hydrant/nozzle. The pump head shall be calculated based on the terminal pressure of 7 barG + static head + friction losses. Design Flows: EPC contractor shall calculate the fire pump flow rate considering the worst-case scenario, but shall not be less than 1000 gpm. Fire Pump shall be FM-approved and UL-listed. Large Electric Motors: In case of a large pump room, partially open and containing lubricants/oils, install both local deluge zones (for fuel fires) and gaseous total-flooding for electrical enclosures. Systems must be zoned to avoid water flooding live switchgear. 2. For the offices at the Ultimate Disposal Station, the Contractor shall design and provide an Automatic wet-pipe sprinkler system (light hazard).

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		Also clarify whether a fire water tank is required and, if so, the required capacity. 3) Whether gas-based fire-extinguishing systems are required in specific rooms such as the control room or substation. If so, kindly identify the applicable rooms. 4) Whether a potable water supply system is required. If yes, please provide pump parameters, tank size, and the source of domestic water. 5) Whether integrated sewage treatment facilities are required at the station and clarify the method for domestic sewage disposal.	3. For the electrical building and pump room, where electrical equipment are installed, at the Ultimate Disposal Station, the Contractor shall design and provide a clean gas-based fire suppression system e.g. Inergen (IG-541) or FM200. For the 23 intake locations, the Contractor shall provide powder-based fire extinguishers. 4. The Contractor shall ensure the supply of potable water at all intake locations and at the ultimate disposal station, including the new building. The primary source shall be the WASA-L water supply, if available in the vicinity. If WASA-L supply is not available, groundwater shall be used as the alternative source, and the Contractor shall install a borehole pump with all associated works. The Contractor shall design the water supply system for the mentioned locations in accordance with WASA-L criteria. 5. There is no requirement to treat the domestic sewage before disposal into the sewerage system at the intake locations and the ultimate disposal station.
11.	Volume 2, Part II, Section 6, Employer's Requirement Sub-Clause 5.1 General Design Obligations	Penstock Gate Valves Operation method: Manual, rising stem in 23 lift station and ultimate disposal station. Please clarify whether motorized actuators are required for Penstock Gate Valves. If required, kindly specify which Penstock Gate Valves in the 23 lift stations and the Ultimate Disposal Station are to be equipped with motor-driven mechanisms.	All penstocks shall be provided with hand-operated gearboxes, designed to allow full opening and closing by a single operator under all operating conditions.
12.	Volume 2, Part II, Section 6, Employer's Requirement 4.1.12. Detailed Description of Scope of Construction Works by the EPC Contractor	xx) Construction of Office Building (Drawing No. LW-TD-401 through Drawing No. LW-TD-403) Page 6-67. Please provide the quantities and specifications of all facilities to be provided within the Office Building.	Please refer to the reply to Sr. No. 44 and 45 of the Clarifications and Minutes of Meeting of 1st Pre-Tender Meeting, that may be accessed on the WASA Lahore website under LWWMP Tab.

第2卷，第1部分，第1	13. 6节，雇主要求 分款6.6 为员工和劳工提供的设施 第6-397页	根据合同条件第6.6款，承包商应为雇主人员提供的设施如下所述： 承包商应确保为雇主提供以下设施： ☐ 配备全套家具和空调的现场办公室，以及休息室和为雇主及PMC现场员工提供的所有办公用品/设施。 ☐ 在办公室安排食品和饮料 ☐ 提供适当的电子及纸质记录维护 ☐ 配备多媒体墙和用于演示项目的交互式3-D模型的设计办公室及委员会/会议室 配备多媒体墙和用于演示的项目交互式3-D模型。 ☐ 提供所有文具、彩色及黑白& ☐ 为有子女的员工安排日间托儿服务。 ☐ 办公区域应配备安保设施、停车区、摄像监控等，确保安全。 ☐ 为雇主的员工提供最新的办公设备 （即配备正版软件和Windows系统的办公计算机/笔记本电脑，以及用于记录现场活动和投诉或跟踪工程进度等的手机和平板电脑） ☐ 提供合理的交通安排 供雇主和PMC员工使用。 ☐ 安排现场附近的员工住宿 以便利现场监督和检查。 ☐ 安排常规及高级别 代表团对工程进行检查，包括破土和开工仪式。 ☐ 承包商应建立媒体联络 安排，以定期向媒体提供信息。 ☐ 安排现场附近的员工住宿 以便利现场监督和检查。 ☐ 安排常规及高级别 代表团对工程进行检查，包括破土	根据合同条件第6.6款，承包商应确保最好在每个活动现场设置现场办公室；在任何情况下，指定现场位置的现场办公室最少不得少于三个（03）个。集装箱应分别供雇主、PMC和承包商等使用。这些办公室应按照雇主要求（ER）中规定的设施、规格和标准提供。
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Sr #	Reference Section	Description	Employer's Response
		breaking & inauguration event. • Facilitate preparation of record and evidences for accountability agencies. • The Contractor shall bear all such costs to be included in bid price. Please specify the number and type of offices to be provided for the Employer, along with the required facilities and specifications.	
14.	Volume 2, Part II, Section 6, Employer's Requirement iii) Component 1: Laying of Trunk Sewers Through Micro-tunneling, Construction of Sewer Appurtenant Structures, and Installation of E&M Equipment Page 6-28	The Contractor shall ensure Manufacturing, Procurement, Delivery and Deployment of at least six number of (06) New MTBMs (OEM Certified) at site with all accessories/ equipment's including but not limited to RT Cranes, dump truck trolleys, trailer, excavator and their power supplying generators with trained O&M professionals and adequate spares readily available at site. The MTBM Machines are recommended to be available and operating at site before 6-months of the commencement of work to ensure timely completion. In no case it shall be delayed more than 8 months for MTBM. These Machines shall be capable to complete the project within the project time without major overhauls during the construction period: Diameter 1.050 m: Two (2) machines Diameter 1.500 m: One (1) machine Diameter 2.400 m: Two (2) machines Diameter 3.500 m: One (1) machine According to the clause requiring the Contractor to procure six new MTBM machines, please clarify the ownership of this equipment upon completion of the Project.	The MTBM machines required for execution of the said Works shall remain the property of the Contractor upon successful completion of the Project.
15.	Volume 2, Part II, Section 6, Employer's Requirement, 4.1.3. THE PROJECT BRIEF Page 6-6	Sewage conveyance /transport system consists of Line-A, Line-B, Line-C and their laterals which consist of about 19.0 km long Trunk Sewers and 13.36 km Lateral/branch and will be laid by micro-tunneling and pipe jacking technique with varying diameter of pipes from 1,050 mm, 1,500 mm, 2,400	Please refer to the reply to Sr. No. 09 of the Clarifications and Minutes of Meeting of 2 nd Pre-Tender Meeting, that may be accessed on the WASA Lahore website under LWWMP Tab.

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		mm and 3,500 mm. Based on the Employer's Requirements and Drawings, the 2400 mm and 3500 mm diameter pipelines are to be installed using micro-tunnelling and pipe-jacking. Please clarify whether the Tenderer may propose an alternative trenchless technology in lieu of the specified method.	
16.	Volume 2, Part II, Section 6, Employer's Requirement	Please clarify the Employer's position regarding the sourcing of major construction materials such as cement and steel, particularly in relation to locally available materials versus imported materials.	It is to clarify that the sourcing of construction materials shall remain the Contractor's responsibility, subject to compliance with the technical specifications and applicable laws and regulations and Employer's consent. In this regard, Contractors are advised to carefully consider the time-related and fiscal implications of importing materials, including applicable taxes, duties, and other related charges. All commonly available and locally manufactured materials forming part of the Permanent Works, including but not limited to cement, steel, sand, aggregates, and similar construction materials, are required to be procured locally from within Pakistan. Import of such locally available permanent materials shall not be permitted and shall require prior written approval of the Employer, supported by adequate technical justification. It is further clarified that the Employer's liability for payment of taxes and duties shall be limited to permanent plant and equipment (machinery) of Employer's use to be installed under the Project. Such taxes and duties shall be paid by the Contractor at the time of import and shall be reimbursed by the Employer through the Interim Payment Certificates (IPCs), in accordance with the provisions of the Contract.
17.	Volume 2, Part II, Section 6, Employer's Requirement	During the site visit, the Tenderer has identified an additional parcel of land probably owned by Lahore Development Authority (LDA), located in proximity to the Project area and WASA owned land, which appears suitable for use as a pipe manufacturing and casting yard. In this regard, the Tenderer seeks clarification	It is to clarify that in addition to the sites identified in the Tender Documents, the Contractor may utilize any other suitable land, including the land owned by Lahore Development Authority (LDA) identified during the site visit, for pipe manufacturing and associated activities, subject to obtaining all requisite approvals and NOCs from the relevant authorities. In this regard, the updated